Industrial Electronics N5

Module 1

Alternating Current Theory and RLC Circuits

Learning Outcomes

At the end of this module, learners will be able to:

- Draw labelled circuit diagrams inclusive of the input and output waveforms for both the RC and RL integrators and RC and RL differentiators
- Explain what is meant by the terms RC differentiator, RC integrator, RL differentiator and RL integrator
- Calculate for both RC and RL circuits;
    - Input and output voltage
    - Capacitance of the capacitor
    - Current
    - Resistance
    - Rate of change of input and output voltage
    - Time constant
- Draw the frequency response of each (RC and RL circuits)
- Explain the term capacitive coupling
- List advantages and the disadvantages of coupling circuit
- Expain the term harmonics
- Draw neatly labelled square waveform and its harmonics
- Analyse RLC circuits phasor diagrams to calculate the following by applying complex numbers in series or parallel circuits;
    - Voltage
    - Current
    - Total impedance
    - Total power
    - Power factor
    - Phasor angle
    - Inductive reactance
    - Capacitive reactance
- Calculate the following values for a series or parallel circuit operating under resonance circuit;
    - Resonant frequency
    - Dynamic impedance
    - Q-factor
    - Bandwidth
    - Reactances

- Indicate the difference between high and low selectivity and bandwidth by means of a neatly labelled amplitude response curve versus frequency response curve

Introduction

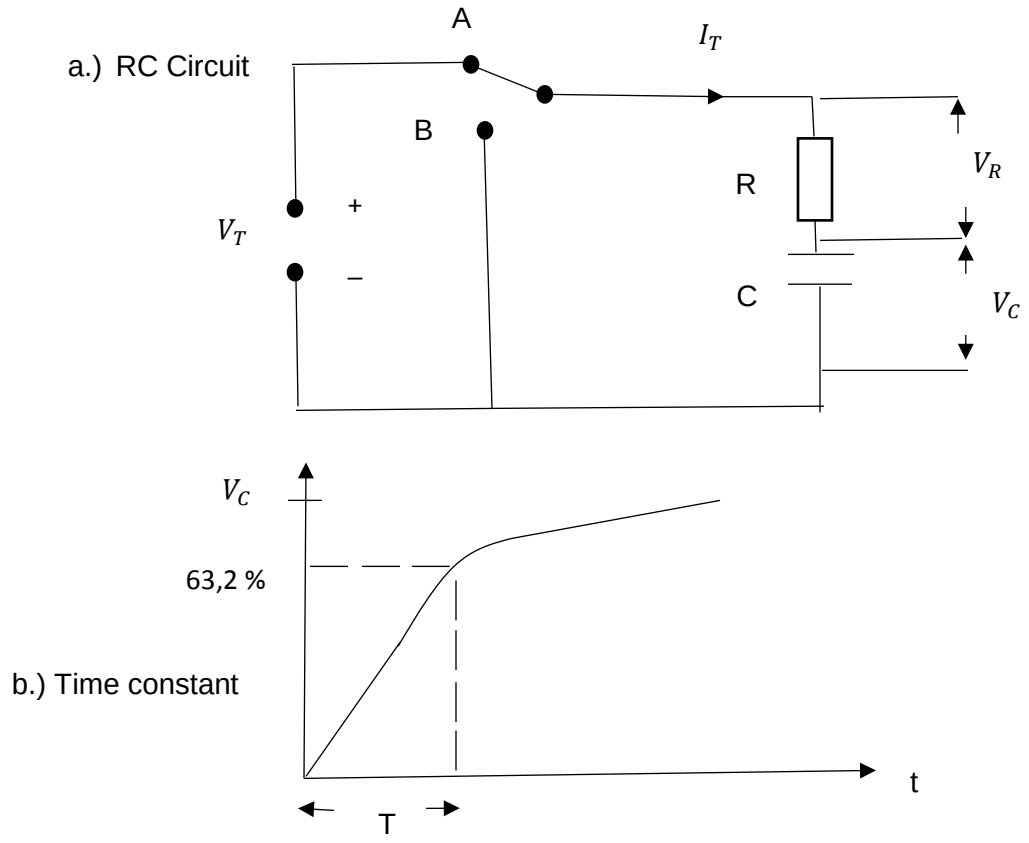
The time constant of an energy-storing circuit can be defined as the time taken for the circuit to reach Steady-state conditions after a disturbance assuming a constant rate of change in the charging or discharging current. ***Every electronic technician must have background knowledge.***

1.1 THE TIME CONSTANT OF AN RC CIRCUIT

A Capacitor can store an electric charge. The following factors can influence its capacitance:

- ***Area*** of Plates
- ***Distance*** between the plates
- The ***Dielectric*** constant

The rate of charge on a capacitor is determined by its capacitance and the resistance in the circuit. The voltage drop across the capacitor tends to the same value as that of the Voltage source, looking at FIGURE 1.1



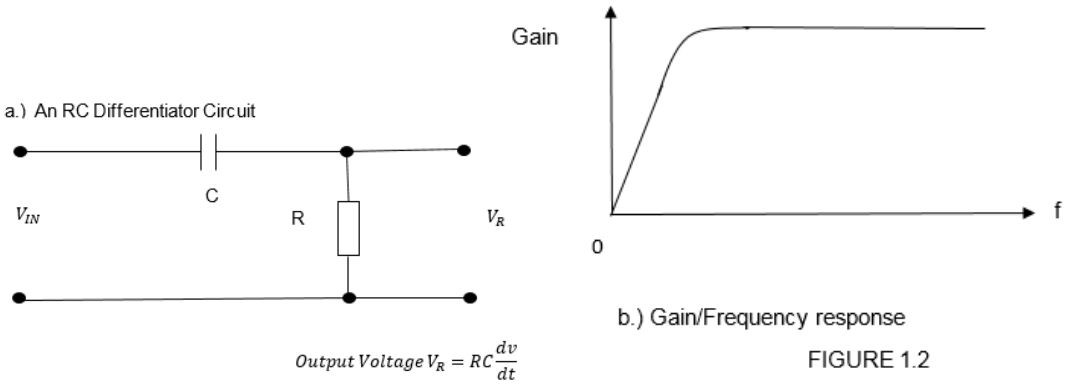
The Capacitor is initially discharged and switched from position B to position A at time t=0, (V_c) will be equivalent to 0 and (V_R) = (V_T). At this moment the total current I_T will have a maximum value. As soon as t>0, V_R and I_C will decrease while V_c increases. When the capacitor voltage $V_c = V_T, I_T = 0$ and $V_R = 0$.

FIGURE 1.1

The time constant of an RC circuit is the time the capacitor takes to charge from 0% to 63,2% of the applied voltage and is calculated from $T = RC\ seconds$.

1.2 THE RC DIFFERENTIATOR

A primary ***RC Differentiator*** consists of an RC circuit of which the output is attained from the resistor. This circuit is also known as a ***high pass filter*** as the capacitor is in series with the output, posing a ***high reactance at low-frequency signals and passing through high-frequency signals.*** This gives the circuit good application in producing ***triggers or spiked pulses in timing circuit applications***. On the circuit below, the output is taken from the resistor, hence why it is called the high pass filter, the capacitor forms the input component. The output response tends to be rectangular unlike that of the integrator being triangular.



$Output\ Voltage\ V_R = RC\frac{dv}{dt}$

FIGURE 1.2

1.2.1 RC DIFFERENTIATOR RULES

The circuit's behavior follows mathematical exponential rule and the equations used are derived using Kirchhoff's laws as follows during the charge and discharge.

DURING THE CHARGE CYCLE;

During the **Charge** cycle from **Kirchhoff's** Voltage Law, we obtain the following:

$Volts\ \ where\ ;\ \ \tau = RC\ seconds\ and\ t = instataneous\ time$

$V_{R\ =\ V_S\left(e^{\frac{-t}{T}}\right)}\ voltage\ across\ the\ resistor$

$V_C = V_S\ (1 - e^{-\frac{t}{T}})\quad voltage\ across\ the\ capacitor\ in\ volts$

$i = I_S\left(e^{\frac{-t}{T}}\right)\qquad Ampere\ instataneuos\ current$

$\omega = 0.5\ C.V^2\ energy\ stored\ by\ the\ capacitor\ when\ it\ is\ fully\ charged\ in\ joules$

$Q = CV\ \ capacitor\ chrage\ in\ coulombs$

$$V_c = \frac{V_s}{\tau}$$

An effective differentiator is obtained when the RC time constant is smaller than the applied square wave mark period. The voltage V_R is derived from the following:

$$V_C = \frac{1}{C}\int idt$$

$$Therefore\ i = (C)\left(\frac{dV_C}{dt}\right)$$

$$V_R = I \times R\ (from\ Ohms\ Law) = (RC)\left(\frac{dV_c}{dt}\right)$$

Since the RC constant is small, $X_C \gg R(X_C\ is\ large\ at\ low\ frequencies)$. Therefore, the total supply voltage will perform across the capacitor almost in its totality, which means that:

$$V_C = V_T\ \ and\ \ V_R = (RC)\left(\frac{dV}{dt}\right)$$

The Voltage V_R is proportional to the differential coefficient of the supply voltage. FIGURE 1.3 depicts the supply and output waveforms with the **RC constant smaller than the marking period of the applied square wave.**

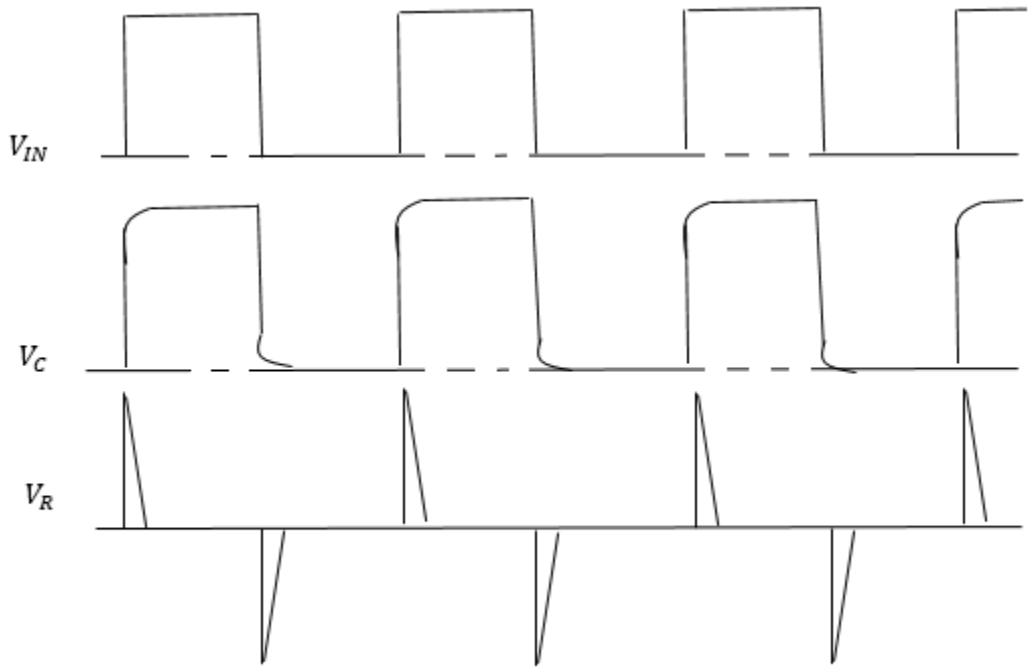


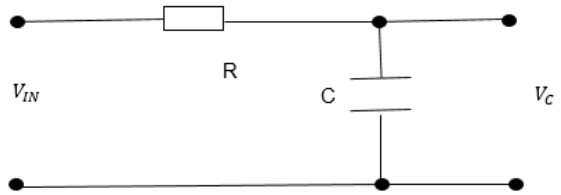
FIGURE 1.3 Waveforms of the Differentiator

1.3 THE RC INTEGRATOR

The RC integrator is attained as soon as the output is taken from the capacitor and the RC constant is much larger than the marking period of the applied square waveform. Therefore, this circuit is a ***low pass filter*** as the capacitor is parallel to the load, short circuits high-***frequency signals.***

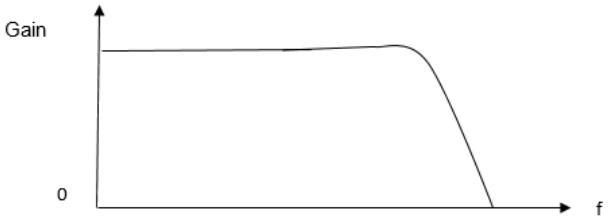
When a step input is applied on an RC integrator, there is a transitional period during which the voltages and currents change from former values to new values after this change. This transitional change is called a ***transient*** and the following will then look closely to a single pulse response applied to the circuit.

RC integrators produce triangular waves from square wave input, they are widely used in ***analog computers***, ***wave shaping circuits*** and ***A/D converters***.



$V_C = \frac{1}{RC}\int V_I\, dt$

a.) An RC integrator circuit



b.) Gain/Frequency response

FIGURE 1.4

If the values of R and C are very large: R>>X_C (X_C is small at high frequencies)

Therefore $i = \frac{V}{R}$, $V_C = \frac{1}{C}\int idt$

$$V_C = \frac{1}{RC}\int Vdt$$

When a step input voltage is applied across the resistor, the capacitor will then begin to charge from zero (assuming initial conditions), pass the 63,2% mark value and approaches a steady state condition. ***This mark value*** is referred to as a time constant as defined previously, ***steady state*** condition is a point during the charge and or discharge where the voltage value in a capacitor remains constant.

FIGURE 1.5 displays the supply and output waveforms of an integrator when an **RC constant is greater than the marking period of the applied square waveform.**

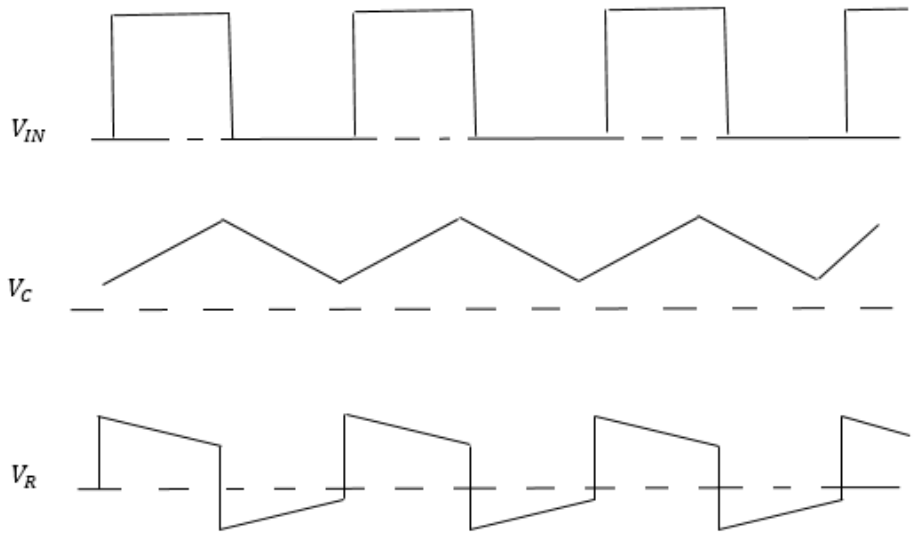


FIGURE 1.5 Waveforms of the integrator

1.3.1 RC INTEGRATOR RULES

The circuit's behavior follows mathematical exponential rule and the equations used are derived using Kirchhoff's laws as follows during the charge and discharge.

DURING THE CHARGE CYCLE;

During the **Charge** cycle from **Kirchhoff's** Voltage Law, we obtain the following:

$Volts\ \ where\ ;\ \ T = RC\ seconds\ and\ t = instataneous\ time$

$V_{R\ =\ V_S\left(e^{\frac{-t}{T}}\right)}\ voltage\ across\ the\ resistor$

$V_C = V_S\ (1 - e^{-\frac{t}{T}})\quad voltage\ across\ the\ capacitor\ in\ volts$

$i = I_S\left(e^{\frac{-t}{T}}\right)\qquad Ampere\ instataneuos\ current$

$\omega = 0.5\ C.V^2\ energy\ stored\ by\ the\ capacitor\ when\ it\ is\ fully\ charged\ in\ joules$

$\mathrm{V_C} = \frac{\mathrm{V_S}}{\mathrm{T}}\ rate\ of\ change\ of\ capacitor\ voltage\ in\ Volts/second$

Example 1.1

A 1µF capacitor is connected in series with a 2.2K Ω resistor. Determine the following should a step voltage of 25 volts with a pulse duration of 5 ms be applied to the circuit mentioned.

a) Time constant
b) The voltage that the capacitor will charge to after one-time constant
c) The voltage across the resistor after one-time constant

d) The initial charge current
e) The rate of change of capacitor voltage
f) Energy stored in the capacitor when fully charged


Solution

a.) $T = R.C = (2.2X\ 10^3)(1X10^{-6}) = 2.2\ ms$

b.) $Vc = Vs\left(1 - e^{-\frac{t}{T}}\right)$

$= 25\left(1 - e^{-\frac{5\times10^{-3}}{2.2\times10^{-3}}}\right) = 22.42\ V$

c.) $V_{R\ =\ V_S\left(e^{\frac{-t}{T}}\right)}$

$= 25\left(e^{-\frac{5\times10^{-3}}{2.2\times10^{-3}}}\right) = 2.58\ V$

d.) $I = \frac{V_S}{R} = \frac{25}{2.2\times10^3} = 11.36mA$

e.) $\mathrm{V_C} = \frac{\mathrm{V_S}}{\mathrm{T}} = \frac{\mathbf{25}}{\mathbf{2.2\times10^3}} = 0.0114\ \mathrm{V/s}$

f.) $\omega = 0.5\ C.V^2 = (0.5)(1X10^{-6})(25^2) = 0.003J$

1.4 RL INTEGRATOR CIRCUIT

An ***inductor*** resists the change in current; it will ***block high-frequency signals and pass DC or low-frequency signals.*** This property of the indictor can thereby be ***used to block unwanted AC signals*** like in a DC power supply or a low pass filter. The inductor is connected in series and can be known as a choke. If an inductor or a capacitor relates to a load in parallel, they will have the ***opposite effect***. The capacitor will pass DC signals to the load while the inductor lets in AC signals.

The time constant of an RL circuit is defined as the time the circuit takes to reach a value of 63.2% of the total current value. This can be expressed as the time taken for voltages and currents to change by 63 % of the total change and is indicated by **T= L / R** seconds.

The RL circuit in FIGURE 1.4 can be used to study the rise and fall of current and voltage in an inductive circuit during the charge and discharge of the inductor

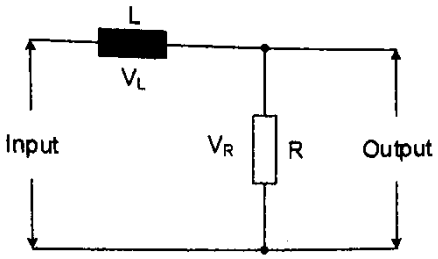
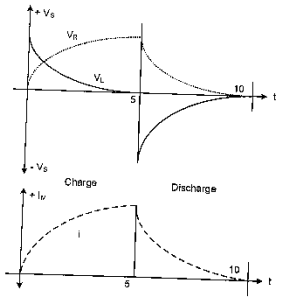


FIGURE 1.4



1.4.1 RL INTEGRATOR RULES

During the **charge** cycle from Kirchhoff's Voltage Law, we obtain the following:

$V_R = V_S\left(1 - e^{-\frac{t}{T}}\right)$ $\quad Volts\ where; T = \frac{L}{R}\ and\ t = instantaneous\ time$

$V_L = V_S\, e^{-t/T}$ $\quad Volts$

$i = I_S\left(1 - e^{-\frac{t}{T}}\right)$ $\quad Amperes$

$W = LI^2 0.5\ JOULES$ $\quad Volts$


Example 1.2

An RL series circuit consisting of an Inductor of 50mH and a resistor of 3 Ω, a 25VDC supply is applied to the RL series circuit.

a.) Calculate the final steady state value of the circuit
b.) Calculate the time constant of the RL series circuit
c.) Calculate the value of the induced emf after 10ms


Solution:

a.) $Steady\ state\ , I = \frac{V}{R} = \frac{25}{3} = 8{,}3\ A$

b.) $T = \frac{L}{R} = \frac{50X10^{-3}}{3} = 0{,}016\ Seconds$

c.) $Vl = V_S\, e^{(-\frac{Rt}{L})}$

$Vl = 25\, e^{(-3\ X\frac{0{,}01}{0{,}05})}$

$Vl = (25)(0{,}5488) = 13{,}72V$


Example 1.3

A coil has a time constant of 17.4 µS and a resistance of 2K3 Ω, when supplied from a 28 V DC source. Determine the time taken for a relay to operate if it pulls the armature in when the current reaches 85 % of its maximum value.


Solution:

$$I = \frac{V}{R} = \frac{28}{2300} = 12{,}2mA$$

$$85\%\ OF\ 12{,}2mA = 10{,}3mA$$

$$I_L = I_S\left(1 - e^{-\frac{t}{T}}\right)$$

$$10{,}3\ X\ 10^{-3} = 12{,}2\ X10^{-3}\left(1 - e^{-\frac{t}{0{,}0000174}}\right)$$

$$t = 32{,}823\ pS$$



Example 1.4

A series RL circuit consists of a 4 Ω, resistor in series with a 2H inductance. Determine:

a.) Instantaneous current after 0,5 seconds
b.) Voltage developed across the resistor after 2 seconds
c.) Maximum energy stored in the magnetic field
d.) Initial rate of current charge



Solution:

a.) $T = \frac{L}{R} = \frac{2}{4} = 0{,}5\ Seconds$

$$I = \frac{V}{R} = \frac{20}{4} = 5A$$

$$i = I_S\left(1 - e^{-\frac{t}{T}}\right) = 5(1 - e^{-\frac{0{,}5}{0{,}5}}) = 3{,}161A$$

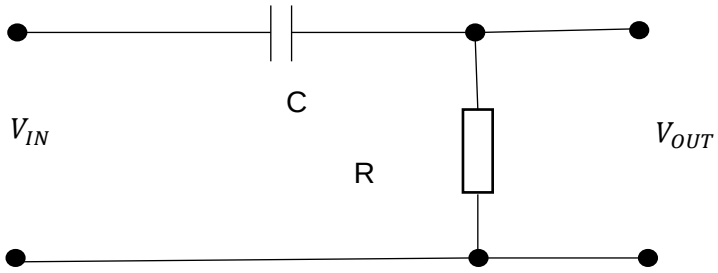
b.) $VR = V_S\left(1 - e^{-\frac{t}{T}}\right) = 20\left(1 - e^{-\frac{2}{0{,}5}}\right) = 19{,}633V$

c.) $Energy = (0{,}5)(L)(i)^2 = (0{,}5)(2)(5)^2 = 25J$

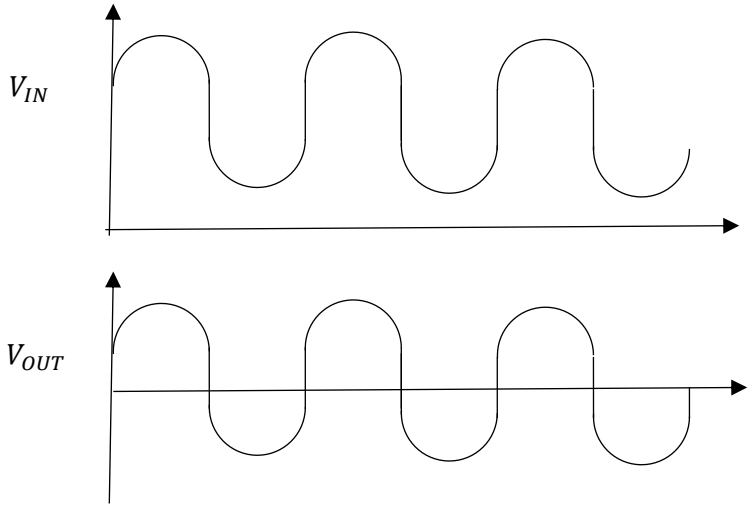
d.) $i = I_S(1 - C^{ut}) = \frac{di}{dt} = \frac{I_S}{T} = \frac{5}{0{,}5} = 10A.Seconds$

1.5 THE CAPACITOR COUPLING CIRCUIT

The output signal of an amplifier normally contains two components, a DC component and an AC component which consists of the information signal. ***The DC Component contains no information and can be isolated from the AC signal with the aid of a coupling capacitor.***



a.) Coupling Circuit



b.) Input and output waveforms

FIGURE 1.5

FIGURE 1.5 shows a coupling capacitor which is like the high pass filter. DC signals can be blocked while AC signals are permitted.

1.6 TRANSIENT BEHAVIOUR IN DC CIRCUITS

We call the response of a circuit immediately after a sudden change the **transient response**, in contrast to the steady state. This phenomenon can take place in RC networks, RL networks and RL networks, irrespective of applying a direct current (DC) source or an alternating current (AC) source.

Terms used in transient circuits:

Familiarise yourself with the following terms related to transient circuits:

Transient means lasting only for a short time. A transient response is a period during which known voltage and/or current values will change from existing values to new values when there is a change in network parameters. Once this time interval has lapsed, the network will be in a steady state.

A **time constant** is the product of the capacitor value and the resistor value (t = R × C) and is the time the network will take to undergo a change of + 63,2% of its initial value. No difference is made between charging and discharging.

The **initial period** for a circuit is the information about the circuit prior to time t = 0, which could include the initial charge or voltage drop across a capacitor, the current flow or the voltage drop across a resistor.

A **transient period or transient response** is the temporary behavior of a system as it adjusts to a new input or disturbance. It includes the initial period of adjustment before the system settles into a stable state.

Steady-state condition is the long-term behavior of a system after it has reached equilibrium following a transient period. In steady state, the system's behavior is constant and predictable under sustained input conditions.

1.7 HARMONICS

A ***sine waveform*** is vital in the testing of electronic circuits. The signals will be sine-shaped if applied across a pure resistive, inductive, or capacitive component. Any other type of waveform will be distorted if an inductance or capacitance exists. Generally, it is necessary to know how a circuit reacts to complex signals (square waveforms). This waveform is applied in digital electronics predominantly as a clock waveform. It is used to determine the distances in ultrasonic appliances and radar systems.

Square waveforms can be easily distressed by unwanted circuit conditions and used for testing circuits. Given that square waveform is composed of such high harmonics, these waveforms can easily observe frequency and phase disturbances. As soon as a perfect square waveform is sent through a faulty circuit, the presence of an error and an indication of the type of error that exists will be designated.

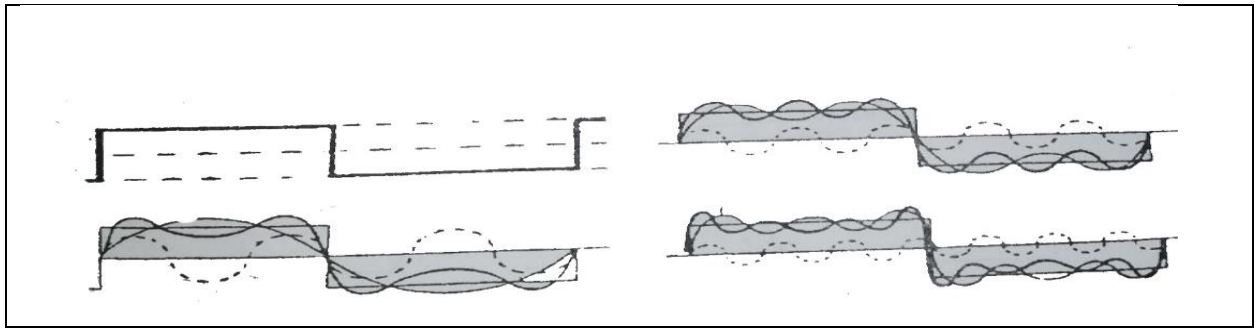


FIGURE 1.6 Harmonics

A square waveform can consist of the sum of:

- A ***fundamental wave***- a sine wave with the same frequency as the square wave

- Several ***harmonics*** – a sinusoidal wave of which the frequencies are integral multiples of the fundamental frequency

Thereby the waveform is composed of the fundamental frequency with the addition of the uneven multiples of the fundamental frequencies (nl 3, 5, 7). The amplitude of any harmonic is indirectly proportional to that harmonic frequency (the 3rd harmonic is one third as powerful as the fundamental, and the fifth harmonic is one fifth as powerful as the fundamental). A shortage in high-frequency response can be observed in the leading and trailing edges of the waveform and low-frequency shortages in the upper horizontal level, as shown in FIGURE 1.7

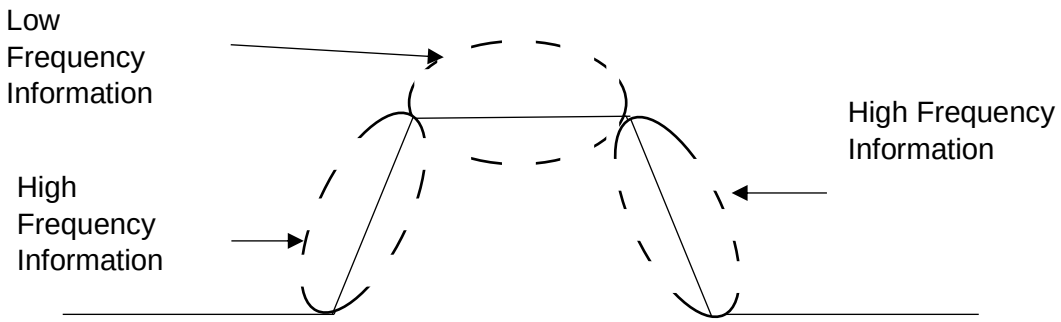


FIGURE 1.7 Low and High frequency shortcomings

Hence, a square wave can indicate the error type, while sine waves often only indicate an error.

Watch the video below for more information on waves: https://youtu.be/vkh8Jfmm3hk

1.8 SERIAL CIRCUITS

In a serial RLC circuit, the total impedance can be calculated by:

$Z = R + J(X_L - X_C)$ Where:

- Z = total impedance
- R= DC resistance
- X_L = Inductive reactance $(2\pi fL)$
- X_C = Capacitive reactance$\left(\frac{1}{2\pi fC}\right)$

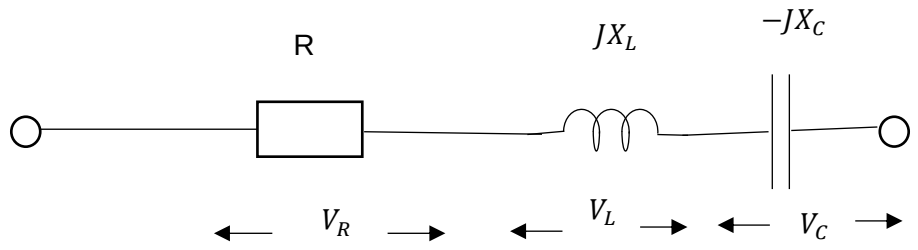


FIGURE 1.8 A serial RLC circuit

The current through the circuit is:

$$I_T = \frac{V_T}{Z_T} \quad and\, V_R = (I_T)(R)$$

$$V_L = I_T(jX_L) \;\; and\, V_C = I_T(-jX_C)$$

Example 1.5

Looking at the circuit in FIGURE 1.9, calculate:

$a.)Z_T$

b.) I_T

$c.)V_R, V_L and\, V_C$, draw the voltage diagram of the circuit.

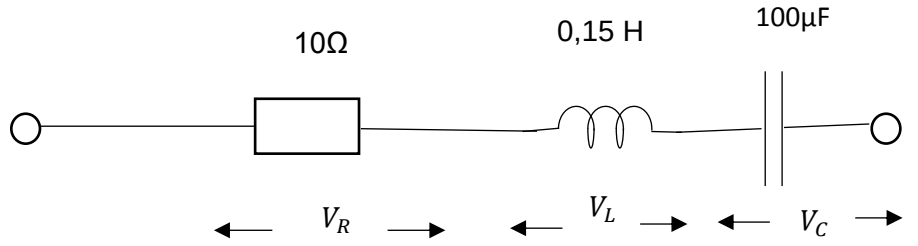


FIGURE 1.9

Solution:

a.) $X_L = 2\pi fL$

$= (2\pi)(50)(0{,}15) = 47{,}124\Omega$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{(2\pi)(50)(100X10^{-6})} = 31{,}831\Omega$$

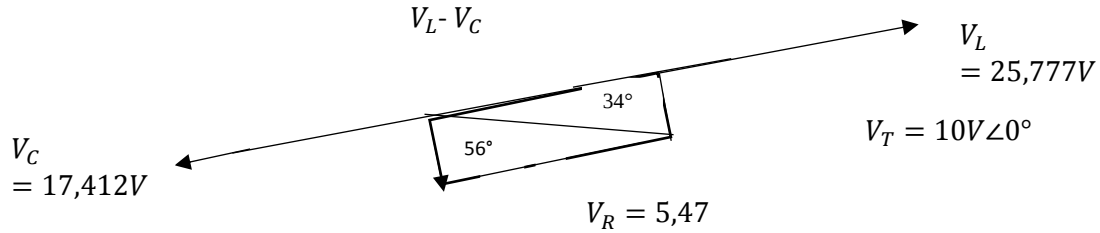
$$Z_T = R + j(X_L - X_C)$$
$$= 10 + j(47{,}124 - 31{,}831)$$
$$= 10 + j15{,}293$$
$$= 18{,}272 \ \angle 56° \ \Omega$$

b.) $I_T = \frac{V_T}{Z_T} = \frac{10/0°}{18{,}272/56°} = 0{,}547\angle -56°$

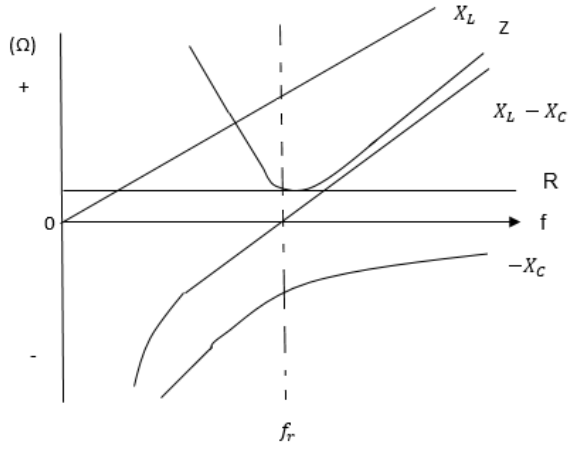
c.) $V_R = (I_T)(R) = (0{,}547\angle -56°)(10\angle 0°) = 5{,}47\angle -56° \ V$

$$V_L = (I_T)(jX_L) = (0{,}547\angle -56°)(47{,}124\angle 90°) = 25{,}7\angle 34°V$$

$$V_C = I_T(-jX_C) = (0{,}547\angle -56°)(31{,}831\angle -90°) = 17{,}412\angle -146°V$$



1.8 RESONANCE IN SERIAL CIRCUITS



If the supply frequency varies, the inductive and capacitive reactance in a serial RLC circuit will also vary.

FIGURE 1.10 Reactance and impedance versus

An increase in frequency will cause an increase in inductive reactance (higher resistance against AC) and a decrease in capacitive reactance(less resistance against AC). In FIGURE

1.10, the variance of X_L ,X_C and Z_T as opposed to frequency is shown. The frequency will also affect the current through the voltage drops across the different components, as seen in FIGURE 1.11 below.

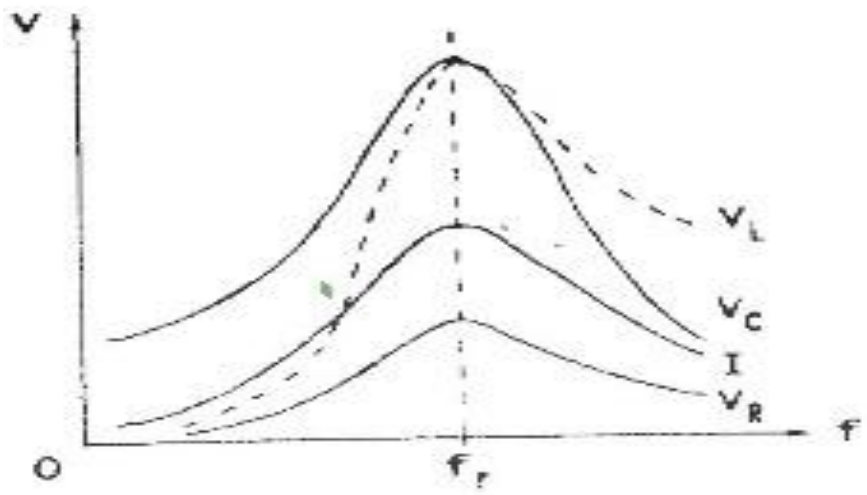


FIGURE 1.11 Voltage and current Vs frequency

At a specific frequency f_r the impedance extends to its minimum value, and the current reaches a maximum value. By that, L and V_C are equal, but they are out of phase with each other. These values can be higher than the voltage supply value, but you must denote a phase difference between the voltage drops. Thereby the circuit is known as a ***resonant*** or a ***tuned circuit.***

The total reactance of a resonant circuit is equal to zero:

$$X_L - X_C = 0$$

$$2\pi fL - \frac{1}{2\pi fC} = 0$$

$$2\pi f = \frac{1}{\sqrt{LC}}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

The total impedance will be:

$$Z = R + j(X_L - X_C)$$

$$Z = R + j(0)$$

$$Z = R$$

Suppose the circuit's resistance is small compared to the inductive and capacitive reactance. The voltage drop across L and C at resonance is larger than the voltage supply. This amplification factor is called the Q-factor and can be calculated by:

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T}$$

$$Q = \frac{(I_T)(X_L)}{(I_T)(R)} = \frac{(I_T)(X_C)}{(I_T)(R)}$$

$$Q = \frac{X_L}{R} = \frac{X_C}{R}$$

$$Q = \frac{2\pi fL}{R} = \frac{1}{2\pi fCR}$$

$$Q = \frac{2\pi\left(\frac{1}{2\pi}\sqrt{LC}\right)(L)}{R} = \frac{1}{2\pi(\frac{1}{2\pi}\sqrt{LC})(CR)}$$

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

If the resistor R in a resonant circuit is adjustable, the current I_T will be adjustable. FIGURE 1.12 depicts the effect on the current for the different values of the resistor. The value of Q is directly proportional to the value of the current. A high Q offers a sharper peak and will discriminate in favor of frequencies close to f_r. This acuteness is known as ***selectivity*** (S = 1/Q). The bandwidth is used to describe the selectivity degree and is defined as the frequency spectrum in which the current exceeds 0,707 of I_T. [1]

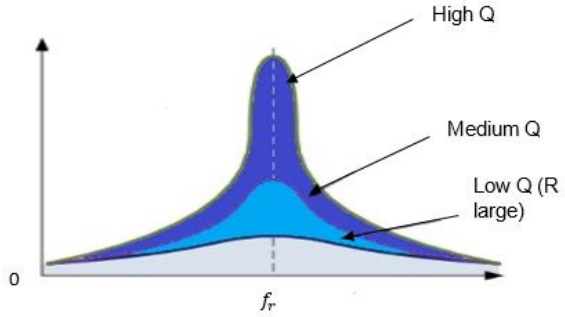


FIGURE 1.12 Selectivity is dependent upon R

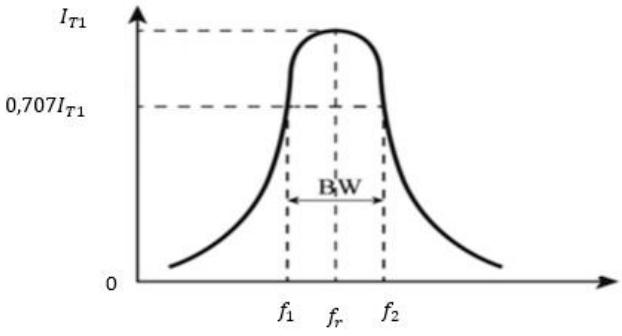


FIGURE 1.13 Bandwidth

The Bandwidth can be calculated as: $Bandwidth = f_2 - f_1$

The Q- Factor can also be calculated by:

$$Q = \frac{Resonant\ frequency}{Bandwidth} = \frac{f_r}{f_2 - f_1}$$

Where $f_{2=}$ Upper half-power frequencies and f_1 = lower half-power frequencies.

In a serial circuit, the Q- Factor is an indication of the voltage amplification as well and be determined by:

$$Q = \frac{V_L}{V_T} \ or \ \frac{V_C}{V_T}$$

Watch the video below fr more information on bandwidth:

https://youtu.be/2GvqQvohP2k



Example 1.6

A series circuit consists of the following:

R = 10Ω, L=0,015H, $V_T = 20V$, $f_r = 1kHz$

a.) Calculate the value of the capacitor required at resonance
b.) Calculate the voltage drops across L and C
c.) Calculate the Q-Factor of the circuit



Solution:

a.) $f_r = \frac{1}{2\pi\sqrt{LC}}$

$$C = \frac{1}{(2\pi f)^2(L)} = \frac{1}{(2\pi\ x\ 1000)^2\ (0{,}015)} = 1{,}68\mu F$$

b.) $I_T = \frac{V_T}{Z_T} = \frac{20}{10} = 2A$

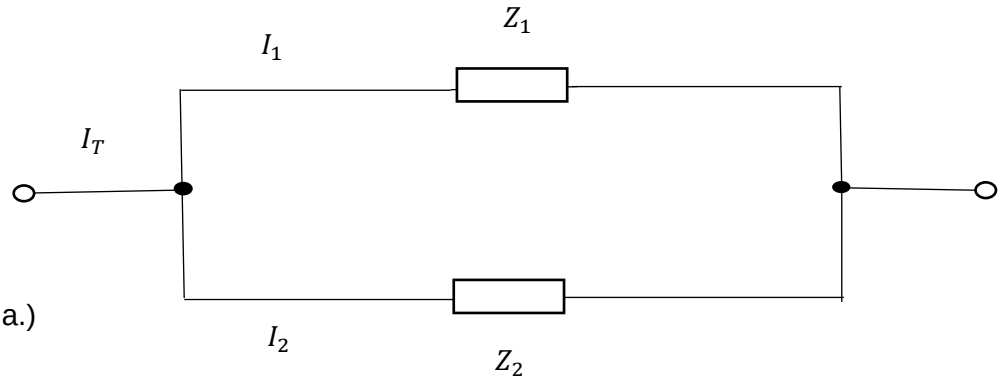
$$V_L = (I_T)(jX_L) = (2)(j2\pi)(1000)(0{,}015) = 188{,}49\angle 90°V$$

$$V_C = (I_T)(-jX_C) = 2\left[(-j)\frac{1}{(2\pi\ X\ 1000\ X\ 1{,}68X10^{-6}}\right] = 188{,}49\angle -90°V$$

c.) $Q - factor = \frac{X_L}{R} = \frac{2\pi\ X\ 1000\ X\ 0{,}015}{10} = 9{,}42$

1.9 PARALLEL CIRCUITS AND SERIAL-PARALLEL CIRCUITS

It can be proved that the total impedance and current values for the parallel circuits below can be calculated as follows:

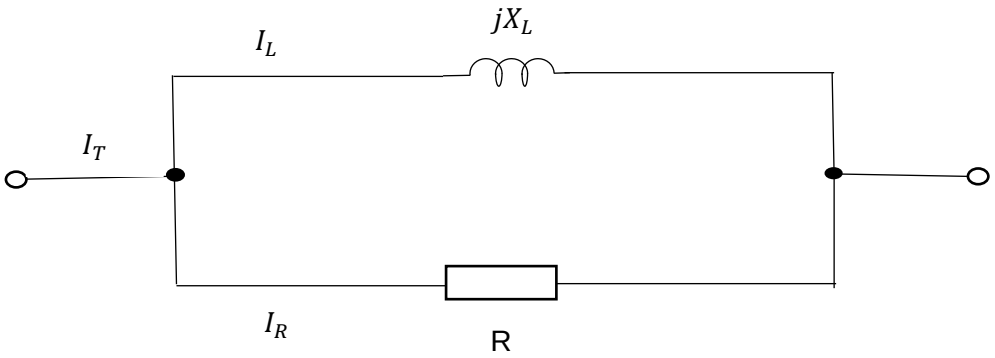


a.)

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$I_T = I_1 + I_2$$

$$= \frac{V}{Z_1} + \frac{V}{Z_2}$$

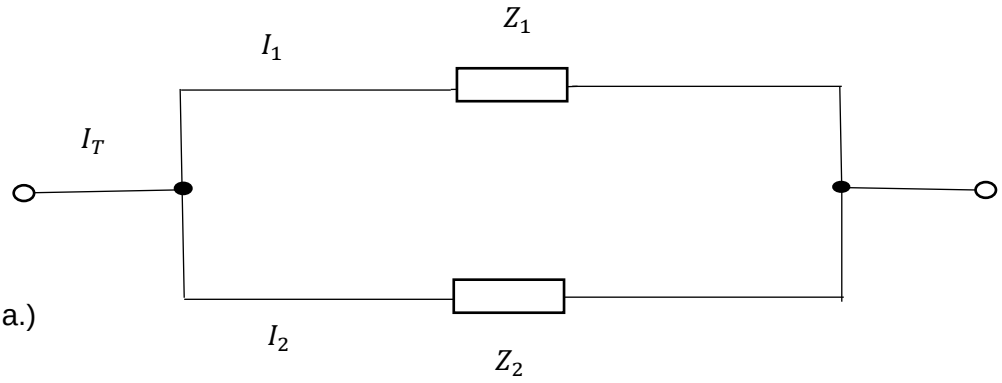


b.)

$$\frac{1}{Z} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L$$

$$= \frac{V}{R} - \frac{jV}{X_L}$$

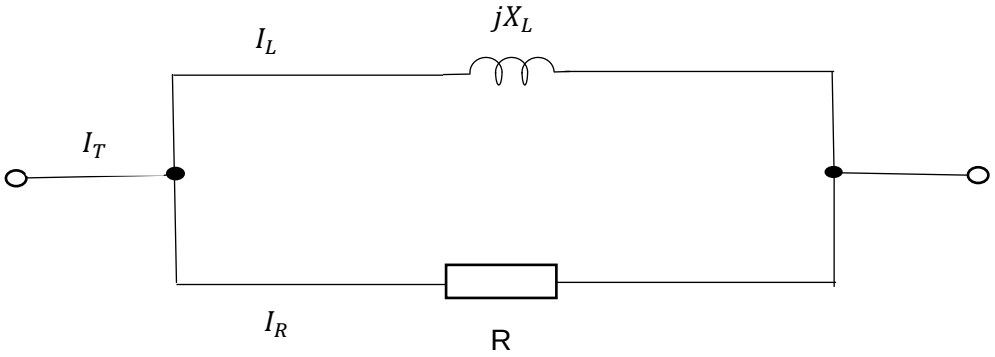


a.)

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$I_T = I_1 + I_2$$

$$= \frac{V}{Z_1} + \frac{V}{Z_2}$$



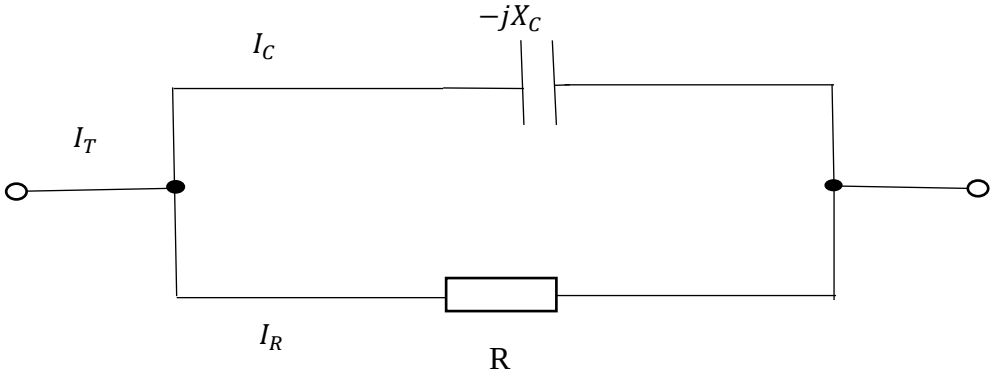
b.)

$$\frac{1}{Z} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L$$

$$= \frac{V}{R} - \frac{jV}{X_L}$$

2

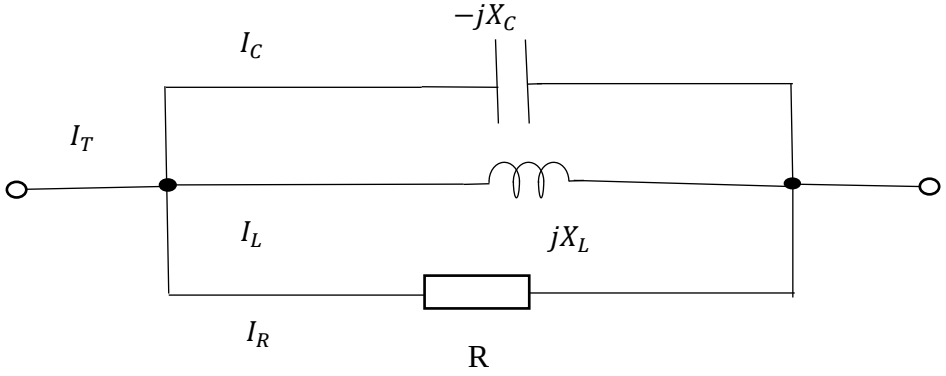


c.)

$$\frac{1}{Z} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C$$

$$= \frac{V}{R} + \frac{jV}{X_C}$$

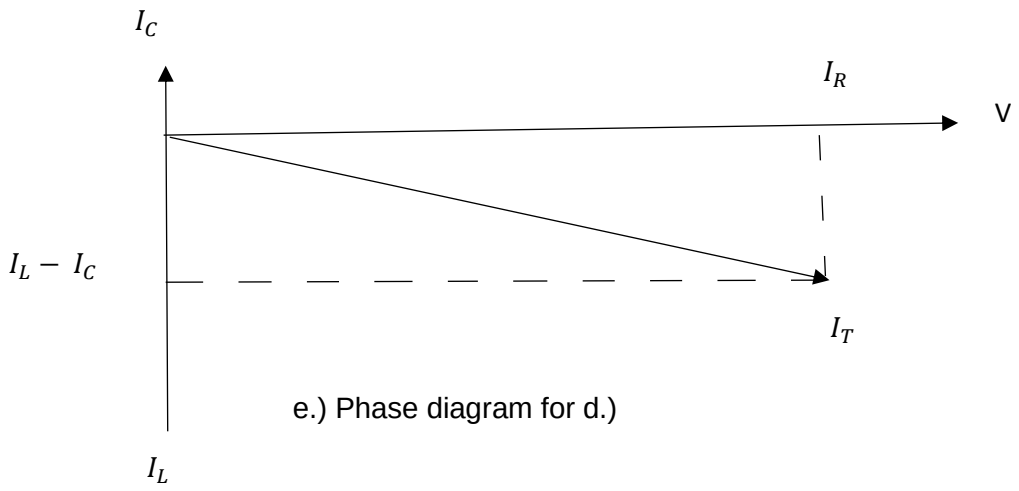


d.)

$$\frac{1}{Z} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = I_R - j(I_L - I_C)$$

$$= \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$



e.) Phase diagram for d.)

FIGURE 1.14 Formulas applicable on parallel circuits

Example 1.7

Referring to FIGURE 1.15, Calculate the following:

a.) The total impedance
b.) The current through every branch
c.) With the aid of a sketch, show the phase diagram for the currents

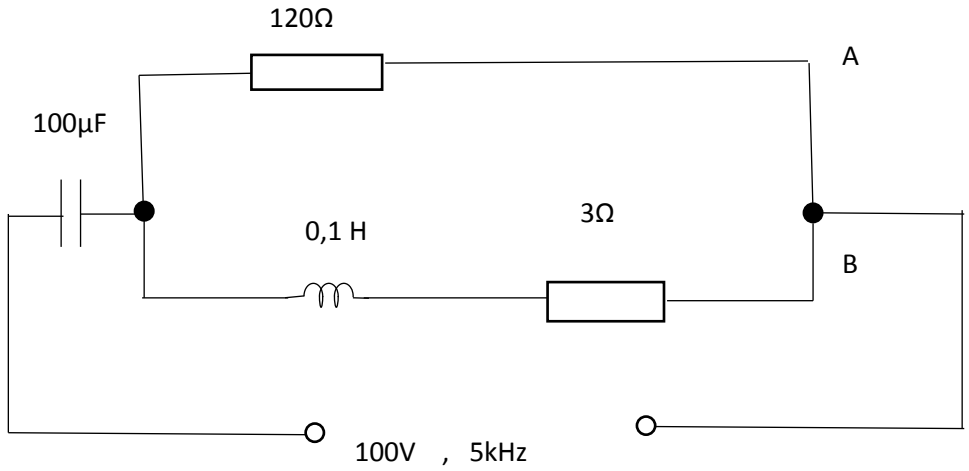


FIGURE 1.15

Solution:

a.) $Z_A = R_1$
$= 120 + j0\Omega$
$= 120\angle 0° \ \Omega$

$Z_B = R_1 + X_L$
$= 3 + (j2\pi)(5000)(0{,}1)$
$= 3 + j3141{,}59\Omega$
$= 3141{,}59\angle 89{,}9°A$

$$Z_{AB} = \frac{(Z_A)(Z_B)}{Z_A + Z_B}$$
$$= \frac{(120 + j0)(3 + j3141{,}59)}{(120 + j0 + 3 + j3141{,}59)}$$
$$= \frac{360 + j3756991{,}12}{123 + j3141{,}59}$$
$$= \frac{376991{,}29\angle 89{,}9°}{3143{,}99\angle 87{,}8°}$$
$= 119{,}9\angle 2{,}1°\Omega$

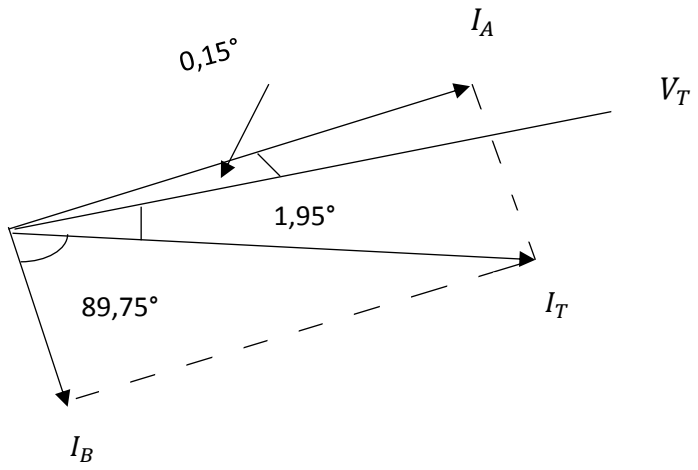
$Z_T = Z_{AB} - jX_C$
$= 119{,}9(COS2{,}1° + jSIN2{,}1°) - j\frac{1}{(2\pi)(5000)(100X10^{-6}}$
$= 119{,}8 + j4{,}39 - j0{,}318$
$= 119{,}8 + j4{,}07\Omega$
$= 119{,}9\angle 1{,}95°\Omega$

b.) $I_T = \frac{V_T}{ZT}$
$$= \frac{100\angle 0°}{119{,}9\angle 1{,}95°}$$
$= 0{,}83\angle -1{,}95°A$

$V_{AB} = (I_T)(Z_{AB}) = (0{,}83\angle -1{,}95°)(119{,}9\angle 2{,}1°) = 100\angle 0{,}15V$

$$I_A = \frac{V_{AB}}{Z_A} = \frac{100\angle 0{,}15°}{120\angle 0°} = 0{,}83\angle 0{,}15°A$$
$I_B = \frac{V_{AB}}{Z_B} = \frac{100\angle 0{,}15°}{3141{,}59\angle 89{,}9°} = 0{,}032\angle -89{,}75°A$

c.)



1.9.1 RESONANCE IN PARALLEL CIRCUITS

The current flowing through the LC circuit can be larger than the total current in a parallel resonant circuit. ***The Q-factor of a parallel resonant circuit is thereby known as the current amplification of this circuit*** and can be calculated as follows:

$$Q = \frac{I_C}{I_T} \; OR \; Q = \frac{I_L}{I_T}$$

It can also be calculated as:

$$Q = tan\theta = \frac{X_L}{R}$$

In a parallel LC circuit, the inductor will have a DC – resistance to an extent (resistance of the conductor), and the circuit can be shown as follows:

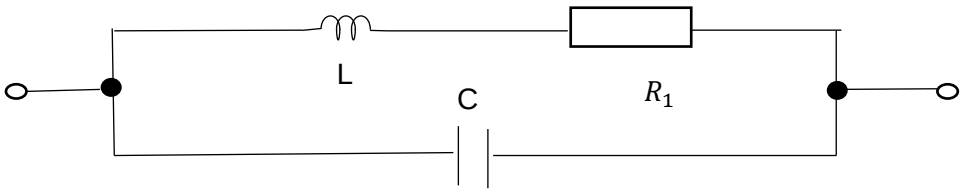


FIGURE 1.16 R_1 *is the DC resistance of* L

At resonance:

$$\frac{1}{Z_T} = \frac{1}{R + jX_L} + \frac{1}{-jX_C}$$

$$= \frac{1}{R + jX_L}\left(\frac{R - jX_L}{R - jX_L}\right) + \frac{1}{-jX_C}\left(\frac{jX_C}{jX_C}\right)$$

$$= \frac{R - jX_L}{R^2 + {X_L}^2} + \frac{jX_C}{{X_C}^2}$$

$$= \frac{R - jX_L}{R^2 + {X_L}^2} + \frac{j}{{X_C}^2}$$

$$= \frac{R}{R^2 + {X_L}^2} - \frac{jX_L}{R^2 + {X_L}^2} + \frac{j}{X_C}$$

$$= \frac{R}{R^2 + {X_L}^2}\ j\left(\frac{1}{X_C} - \frac{X_L}{R^2 + {X_L}^2}\right)$$

Resonance originates when:

$$\frac{1}{X_C} = \frac{X_L}{R^2 + {X_L}^2}$$

$$R^2 + {X_L}^2 = (X_L)(X_C)$$

$$R^2 + (2\pi f)^2 = \frac{L}{C}$$

$$(2\pi f)^2 = \left(\frac{L}{C} - R^2\right)\left(\frac{1}{L^2}\right)$$

$$2\pi f = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$\frac{R^2}{L^2}$ is usually smaller than $\frac{1}{LC}$ because R<<X_L and can thereby be ignored.

Therefore $f = \frac{1}{2\pi\sqrt{LC}}$

The impedance formula of a serial resonant circuit, also known as the dynamic impedance, can be calculated by the following formula: $Z_d = \frac{L}{CR_1}$

Example 1.8

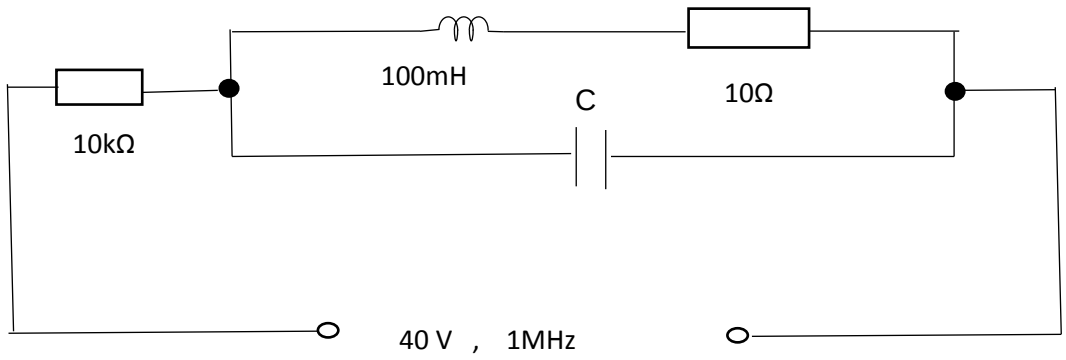


FIGURE 1.17

Referring to FIGURE 1.17, calculate the following:

a.) The value of the capacitor to make the circuit resonate
b.) The dynamic impedance
c.) The Q –factor of the tuned circuit

Solution:

a.) $X_L = 2\pi fL = (2\pi)(1X10^6)(100X10^{-6}) = 628{,}3\Omega$

$f = \dfrac{1}{2\pi\sqrt{LC}}$ $Because\ (R_1 \ll X_L)$

$C = \dfrac{1}{(2\pi f)^2 L} = \dfrac{1}{(2\pi)(1X10^6)(100X10^{-6})} = 253{,}3pF$

b.)

$Z_d = \dfrac{L}{CR_1} = = \dfrac{100X10^{-6}}{(253{,}3X10^{-12})(10)} = 39{,}478k\Omega$

c.) $Q = \dfrac{X_L}{R} = \dfrac{628{,}3}{10} = 62{,}83$

SELF CHECK 1

1.) Which factors determine the charging rate of a capacitor?

2.) Which factors determine the capacitance of the capacitor?

3.) Define the term transient.

4.) Define the term time constant

5.) For an RC integrator with R = 10 ohms, C= 100 micro farad, supply voltage of 20 V and an applied pulse lasting for 1 second. Calculate the voltage across capacitor during charge, (b) the rate of change of capacitor voltage (c) draw a neat waveform of the output

6.) An RL integrator circuit has R = 50 ohms, L = 50 mH, 20V pulse having a width of 1ms applied to the circuit. Calculate (a) the time constant, (b) voltage across the resistor (c) how long will it take for the output voltage to decay to zero?

7.) Why is the Differentiator known as the High pass filter?

8.) Why is the Integrator known as the low pass filter?

9.) Draw a simple gain/frequency response for an integrator and a differentiator.

10.) Show the supply and output waveforms of a Differentiator (with t<mark-period) and an integrator (with t>mark period).

11.) What are the purposes that coupling capacitors are used for?

12.) Why are square waves the most applicable for testing certain electronic circuits?

13.) Low and high-frequency disturbances can be observed from different levels of a square sample waveform. With the aid of a sketch, show the levels involved.

14.) A serial RLC circuit connected to a 250V AC supply consists of 0,15H, A resistor of 10Ω and a capacitor of 100μF. Calculate the total impedance, total current and the Q-factor, given that the supply frequency is 50Hz.

15.) A serial circuit consists of a 12Ω resistor, a 150μH inductor and an adjustable capacitor. This circuit is connected to a 10V,50Hz sine wave; calculate the following:

a.) The value of the capacitor to let the circuit resonate at 50kHz
b.) The total current through the circuit
c.) The voltage drops across each component

d.) The reactance $X_L \; and \; X_C$
e.) The total impedance and the Q-factor
f.) Draw the phase diagram of the voltage drops in this resonant circuit.
g.) Calculate the total impedance
h.) Calculate the voltage drops across the resistor, inductor, and capacitor of a serial circuit with the following values: R=5Ω, X_L=j40Ω, X_C =-j80Ω, $V_T = 30V$
i.) Draw the phasor diagrams of the impedances and voltage drops

16.) An inductor of 5mH is connected in series with an adjustable capacitor at which value must the capacitor be tuned to tune the circuit at 300kHz?

17.) A serial circuit consists of a resistor of 20Ω, an inductor of 10mH, a capacitor of 10μF and a supply voltage of 250V,50Hz.

Calculate:

- The voltage drops across each component and draws the phase diagram
- The value at which the capacitor must be tuned to let the circuit resonate

18.) A capacitor is connected to a 110V,60Hz supply; if the current flow is 250mA, what is the capacitance of the capacitor?

19.) The following 50Hz impedances are given:

$Z_1 = 3 + j10\Omega$, $Z_2 = 5 - j16\Omega$

a.)What is the total impedance of the form if $Z_1$and Z_2 is connected in series?
b.) What is the total impedance of the form if $Z_1$and Z_2 is connected in parallel?

20.) A parallel RL circuit with j$X_L = j15\Omega$ and R=20Ω:

a.) Calculate the current through R and L and the total current with a voltage supply of 20V, 800Hz?
b.) Draw a phase diagram of the currents.

21.) Draw the phase diagram of the currents if a capacitor replaces the inductor in question 16 with a reactance of –j15Ω.

22.) A Coil with a 20milli Henry inductance and a DC resistance of 5Ω is connected to an adjustable capacitor in parallel. The combinations are connected to a 5kΩ resistor in series:

a.) Calculate the resonant frequency with C tuned to 10μF.
b.) Calculate the dynamic impedance.
c.) Calculate the Q-factor of the circuit

23.) A serial RL circuit with values of j31,4Ω and 2Ω is connected across a capacitor in parallel. If the supply voltage is 100V, 500Hz:

a.) Calculate the value of the capacitor at resonance
b.) Calculate the current through each branch.
c.) Calculate Z_d
d.) Calculate the Q-factor.

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